5.11 Convert to binary (Lab 5.1)

LAB ACTIVITY | 5.11.1: Convert to binary (Lab 5.1) | Full screen | 0 / 28

First, download this template and rename it `binary.py`:

- template.py

Write a program that takes in a positive integer as input, and outputs a string of 1's and 0's representing the integer in binary (reversed). For an integer x, the algorithm is:

1. **Set** result to the empty string (i.e., '').
2. **Repeat** as long as x is greater than 0
    1. **Set** result = (x modulo 2) + result (n.b., remainder is either 0 or 1).
    2. **Set** x = x divided by 2 (rounded down to the nearest integer).
3. **Print** result.

Note:

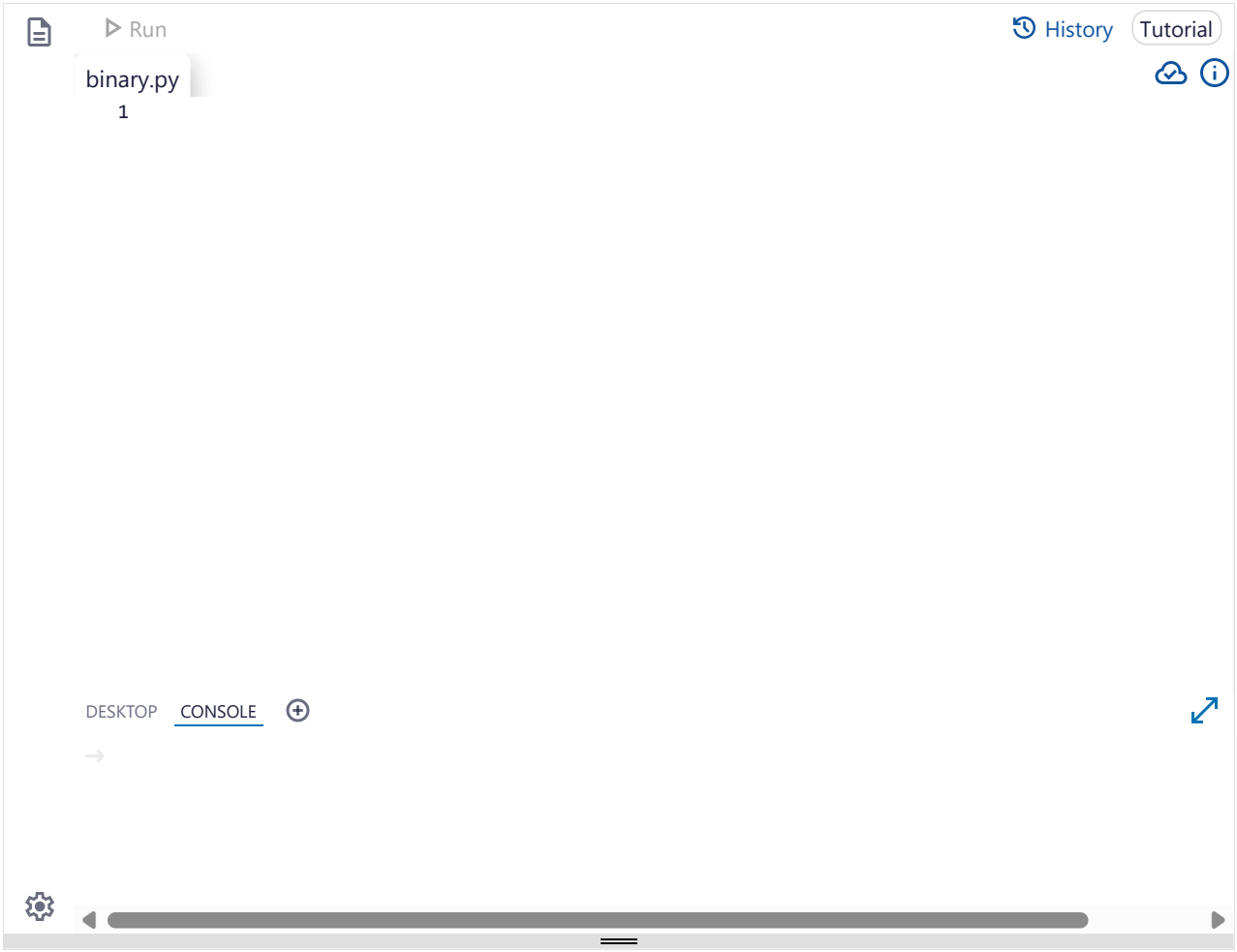
- The above algorithm uses the *accumulator pattern* in that it initializes an empty result value (''), updates that result in the loop, and prints/returns that result at the end. This is a common algorithmic pattern.
- The command that updates the value of `result` using string concatenation (`+`), which requires that you convert the numeric value to a string, i.e., `str(x % 2)`.

E.g., Here is a sample run for the integer 6.

```
integer: 6↵
110
```

*This is based on a ZyLabs example.

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